

XEROX

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Printer thanks joe

Spruce version 11.2 -- spooler version 11.2

File: DSK: KMP; E 12

Creation date: 14 DEC 1984 2046

For: KMP

4 total sheets = 3 pages, 1 copy.

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```
;;; -*- Mode:LISP -*-
```

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#M
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```
(SSTATUS MACRO /:
```

```
  '(LAMBDA NIL
```

```
    ((LAMBDA (SYM)
```

```
      (IF (NOT (SYMBOLP SYM)) (ERROR "Bad keyword" SYM))
```

```
      (IF (EQ (GETCHAR SYM 1.) (QUOTE /:))
```

```
          SYM
```

```
          (SETQ SYM (IMplode (CONS (QUOTE /:) (EXPLODEN SYM)))))
```

```
      (IF (NOT (BOUNDP SYM)) (SET SYM SYM))
```

```
      SYM))
```

```
    (READ))))
```

```
(CONDITION-BIND ((condition handler)
```

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                  (condition handler)
```

```
                  ...))
```

```
  . body)
```

```
(DEFINE-CONDITION name (...args...) (...parent-types...)
```

```
  doc-string
```

```
  . default-handler)
```

```

(DEFINE-CONDITION CONDITION () ()
  "condition"
  NIL)

(DEFINE-CONDITION DEBUGGER-CONDITION () (CONDITION)
  "debugger condition"
  (DBG))

(DEFINE-CONDITION ERROR () (DEBUGGER-CONDITION)
  "error condition")

(DEFINE-CONDITION WRONG-TYPE-ARG (DATA PREDICATE STRING) (ERROR)
  "wrong-type argument condition")

(SIGNAL-PROCEED-CASE ('WRONG-TYPE-ARG :DATA      '3
                                     :PREDICATE #'FLOATP
                                     :STRING    "a floating point number")

  ((:NEW-VALUE      VALUE) VALUE)
  ((:STORE-NEW-VALUE VALUE) (SETQ TEMP VALUE))
  ((:FIX-GIVEN-VALUE)      (SETQ TEMP (FIX TEMP)))))

(DEFMACRO SIGNAL-PROCEED-CASE (CONDITION &BODY DATA)
  (LET ((ARGS (NCONS CONDITION))
        (CASES DATA))
    (DO ()
      ((OR (NULL CASES)
           (NOT (ATOM (CAR CASES)))))
        (PUSH (POP CASES) ARGS)
        (PUSH (POP CASES) ARGS)
        '(SIGNAL-PROCEED-CASE-INTERNAL ,(NREVERSE ARGS) ,@CASES)))

  (DEFMACRO SIGNAL-PROCEED-CASE-INTERNAL (ARGS &BODY CASES)
    (LET ((KEY (GENSYM))
          (TEMP (DO ((L CASES (CDR L))
                    (X NIL))
                    ((NULL L) (MAPCAR #'(LAMBDA (NIL) (GENSYM)) X))
                    (IF (AND (NOT (ATOM (CAAR L)))
                          (> (LENGTH (CDAAR L)) (LENGTH X)))
                        (SETQ X (CDAAR L))))))
          '(MULTIPLE-VALUE-BIND (,KEY ,@TEMP)
            (SIGNAL (MAKE-CONDITION
                     ,@ARGS
                     :PROCEED-TYPES (MAPCAR #'(LAMBDA (CLAUSE)
                                                       (IF (ATOM (CAR CLAUSE))
                                                           (CAR CLAUSE)
                                                           (CAAR CLAUSE)))
                     CASES)))
              (SELECTQ ,KEY
                ,@(MAPCAR #'(LAMBDA (CLAUSE)
                              (COND ((ATOM (CAR CLAUSE))
                                    '((,(CAR CLAUSE)) ,@(CDR CLAUSE)))
                                (T
                                 '((,(CAAR CLAUSE))
                                   ,@(IF (CDAR CLAUSE)
                                         '((LET ,(MAPCAR #'LIST
                                                           (CDAR CLAUSE)
                                                           TEMP)
                                           ,@(CDR CLAUSE)))
                                         (CDR CLAUSE))))))
                              CASES)
                (OTHERWISE (FERROR NIL "Invalid proceed type: ~S" ,KEY))))))

```

```
;;; Local Modes:  
;;; Lisp SIGNAL-PROCEED-CASE Indent:1  
;;; Lisp CONDITION-BIND Indent:1  
;;; End:
```